UNIT 3 · TOPIC 3.3

Protecting Networks: Segmentation

Quiz

Instructions: This quiz has 9 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. Which of the following is a technique used to segment a network?

- ○ A. Installing antivirus software on all endpoint devices
- ○ B. Enabling HTTPS on an organization's public-facing web server
- ○ C. Creating VLANs to separate groups of devices into distinct logical segments
- ○ D. Requiring multi-factor authentication for all user logins

2. A network administrator configures separate VLANs for the HR department and the Engineering department on the same physical switch. What does this configuration primarily accomplish?

- ○ A. It encrypts all traffic traveling between HR and Engineering devices.
- ○ B. It separates the two departments into distinct logical network segments.
- ○ C. It prevents external attackers from reaching either department's devices.
- ○ D. It automatically applies firewall rules to traffic between the two departments.

3. Which of the following best describes a security benefit of network segmentation?

- ○ A. It eliminates the need for encryption because traffic stays within isolated segments.
- ○ B. It improves network performance by reducing the number of nodes in each broadcast domain.
- ○ C. It limits the spread of an attack by containing compromised devices within a single segment.
- ○ D. It prevents insider threats by blocking all employee access to sensitive servers.

4. A hospital network is segmented into three VLANs: one for patient monitoring devices, one for administrative workstations, and one for the billing system. An attacker successfully installs malware on an administrative workstation. Which outcome is most likely as a direct consequence of the segmentation?

- ○ A. The attacker cannot reach patient monitoring devices or the billing system without traversing an inter-VLAN routing control point.
- ○ B. The malware will be automatically detected and removed by the VLAN switch firmware.
- ○ C. The malware will be confined to the workstation's hard drive with no network access at all.
- ○ D. The attacker will gain access to all three VLANs because malware can forge VLAN tags.

5. A company has a flat network where 400 employee workstations, a public web server, and an internal payroll database share the same broadcast domain. The security team's primary concern is a compromised employee workstation reaching the payroll database. Which segmentation approach best addresses this goal?

- ○ A. Move all employees to a single VLAN but increase logging on the payroll database server.
- ○ B. Encrypt all traffic on the flat network using TLS to prevent unauthorized reading of payroll data.

○ C. Separate the web server, employee workstations, and payroll database into distinct VLANs with inter-VLAN traffic controlled by a firewall.

○ D. Apply a uniform strong password policy across all devices so a single compromised credential cannot reach the payroll database.

6. Compared to a flat network, how does VLAN-based segmentation improve network security?

○ A. It enforces 802.1X authentication on all devices, preventing unauthorized users from joining the network.

○ B. It eliminates the need for a firewall because each VLAN functions as its own security perimeter.

○ C. It automatically encrypts intra-VLAN traffic to prevent data interception within a segment.

○ D. It reduces the blast radius of a security incident by limiting which devices can directly communicate with each other.

7. A security architect is comparing two network designs for a corporate campus. Design A uses a single flat network with a hardened perimeter firewall. Design B uses VLAN-based segmentation with per-segment access controls at each inter-VLAN routing point. The architect's primary concern is minimizing the impact of a compromised employee workstation being used by a malicious insider. Which evaluation best supports choosing Design B over Design A?

○ A. Design B is preferred because segmentation eliminates insider threats entirely by preventing all employee-to-server traffic.

○ B. Design A is preferred because a properly configured perimeter firewall is sufficient to contain threats originating from inside the network.

○ C. Design B is preferred because VLAN segmentation forces intra-campus lateral movement through access-controlled routing points, reducing the set of resources reachable from any single compromised device.

○ D. Design A is preferred because VLAN segmentation increases administrative complexity without providing measurable security benefit beyond a perimeter firewall.

8. A network architect proposes layering zero-trust principles on top of existing VLAN boundaries. A colleague argues that VLANs already provide adequate segmentation and zero trust adds unnecessary complexity. Which response most accurately evaluates this disagreement?

○ A. The colleague is correct: VLANs fully enforce zero trust because devices in different VLANs cannot communicate by default.

○ B. The architect is correct because VLANs create coarse-grained perimeters, zero trust adds per-session identity and authorization verification that VLAN boundaries alone cannot enforce.

○ C. The architect is correct because zero trust eliminates the need for VLANs, making network segmentation redundant.

○ D. Both are incorrect: zero trust is an application-layer security model and does not interact with network-layer segmentation.

9. A naval submarine network technician manages three LANs: one for weapons systems, one for official naval operations, and one for crew recreation. The technician must recommend segmentation controls proportional to each LAN's sensitivity level. Which design is most defensible from a security standpoint?

○ A. Merge the weapons and operations LANs into one high-security VLAN and isolate only the recreation LAN, reducing administrative complexity while consolidating the most sensitive systems.

○ B. Connect all three LANs to a central switch without routing restrictions but install an intrusion detection sensor on the weapons LAN only.

○ C. Maintain three fully isolated segments: air-gap or hardware-firewall the weapons LAN from all other traffic, enforce strict access controls between operations and recreation, and treat recreation traffic as untrusted.

○ D. Allow the operations LAN to communicate freely with both the weapons and recreation LANs, since all crew members accessing those LANs hold appropriate clearances.